

1.2. Logic Gates

Logic gates are the fundamental building blocks of digital systems.

Types of logic gates: There are three basic logic gates, namely

- OR gate
- AND gate
- NOT gate

And other logic gates that are derived from these basic gates are:

- NAND gate
- NOR gate
- Exclusive OR gate
- Exclusive NOR gate

1.2.1. NAND gate:

The term NAND gate equivalent to AND gate followed by a NOT gate, implies NOT-AND

Symbol:



Fig. 1.12.

Truth table of 2-input NAND gate.

Input		Output
A	B	$Y = \overline{AB}$
0	0	1
0	1	1
1	0	1
1	1	0

Switching and Circuit Diagram for NAND gate.

B	A	OUT
0	0	1
0	1	1
1	0	1
1	1	0

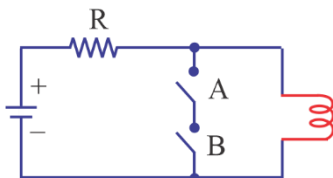


Fig. 1.13.

NAND gate acts as Universal Gate

Logic Gates using only NAND Gates

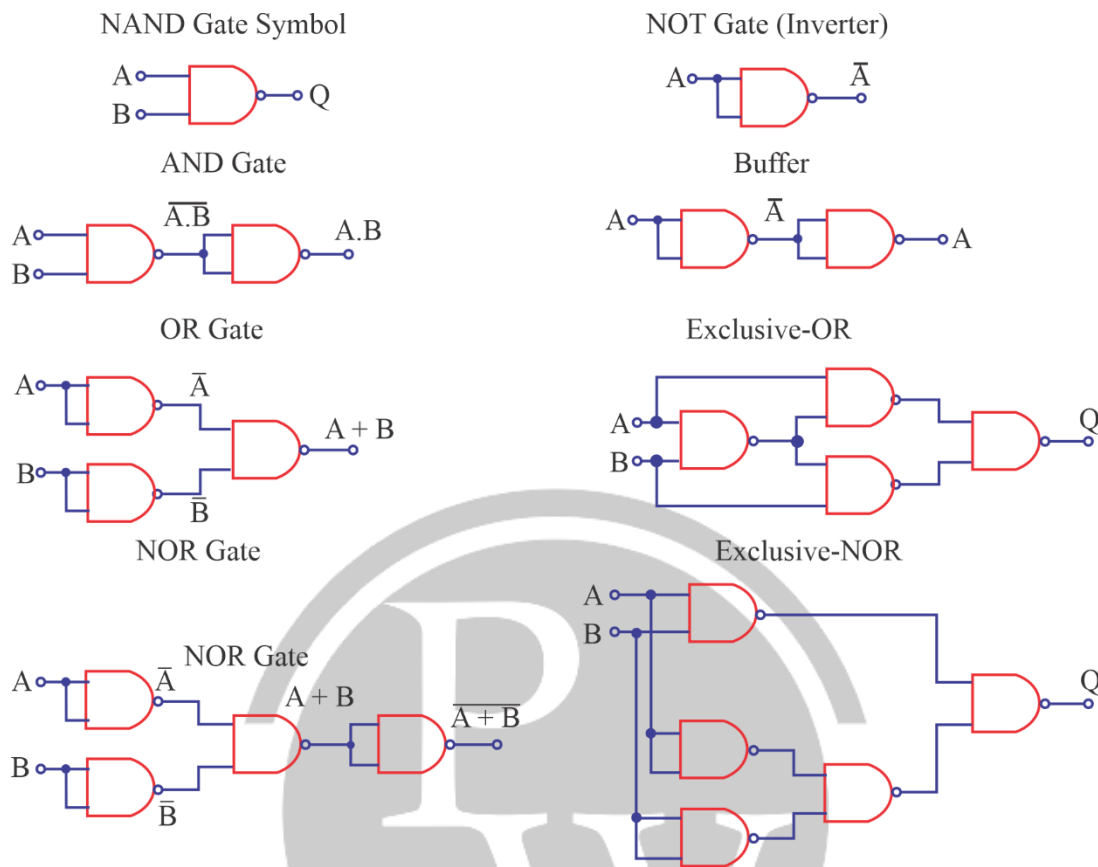


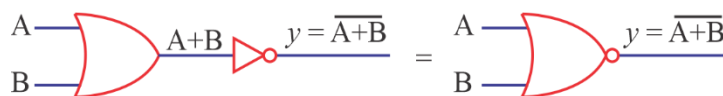
Fig. 1.14.

All the logic gate functions can be created using only NAND gates. Therefore, it is also known as a Universal logic gate.

1.2.2. NOR Gate

A NOR gate is equivalent to OR gate followed by a NOT gate.

Symbol:



Truth Table for 2-input NOR gate

Input		Output
A	B	$Y = \overline{A + B}$
0	0	1
0	1	0
1	0	0
1	1	0

Switching and Circuit Diagram for NOR gate

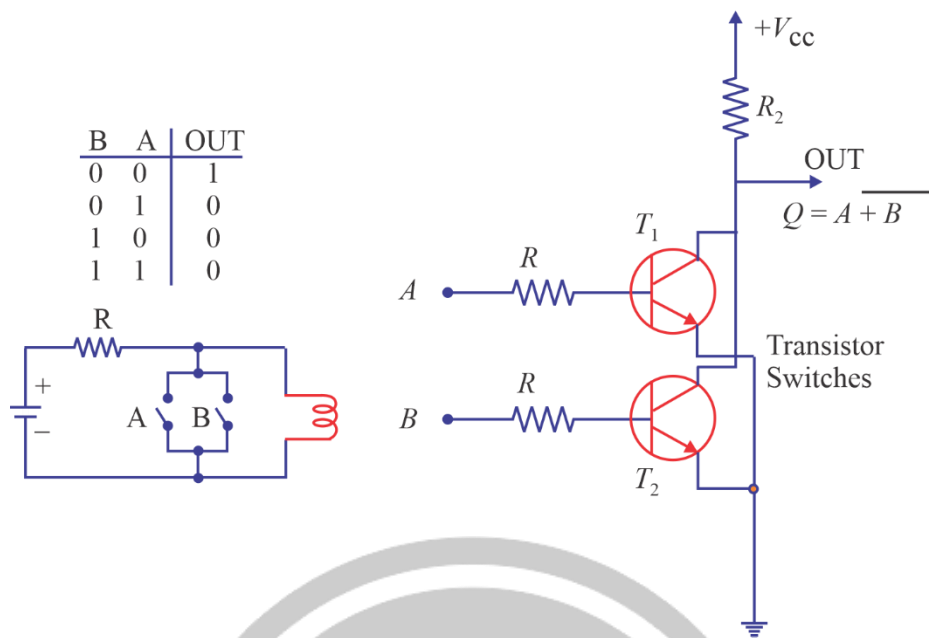


Fig. 1.15.

NOR gate acts as Universal Gate.

Logic Gates using only NOR Gates

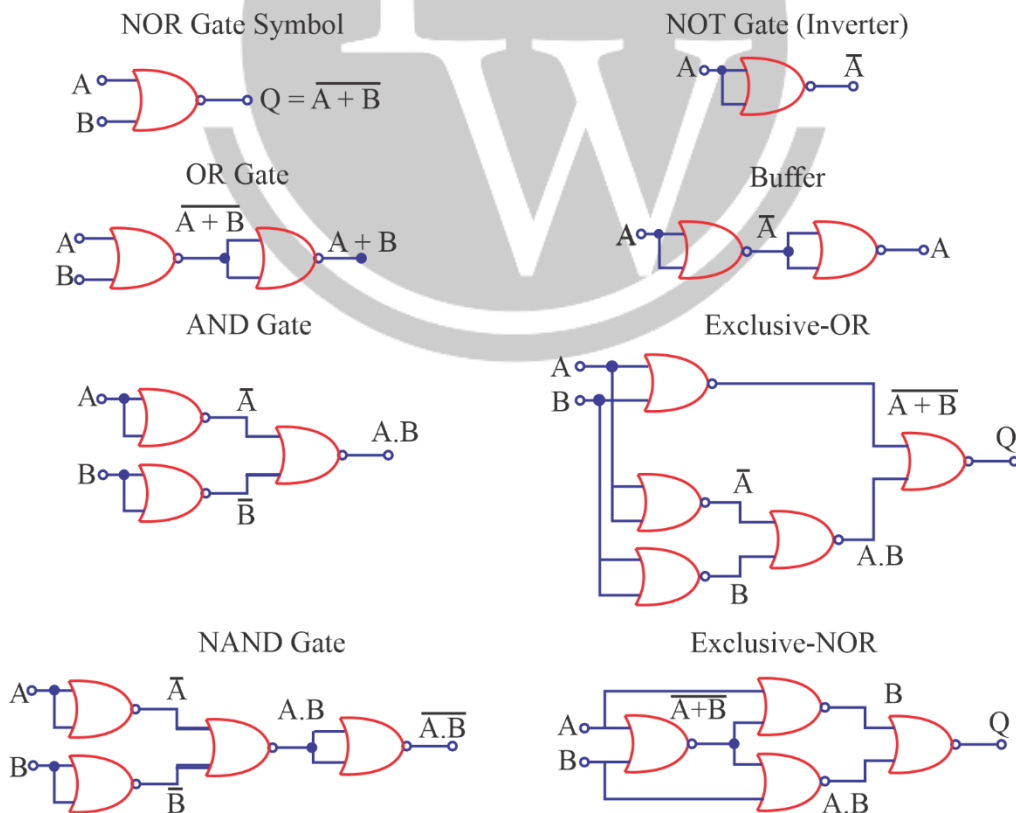
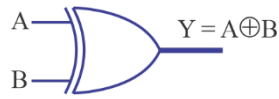


Fig. 1.16.

All the logic gate functions can be created using only NOR gates. Therefore, it is also known as a Universal logic gate.

1.2.3. XOR Gate

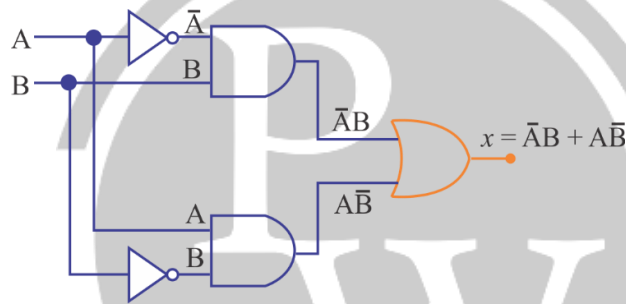
Symbol of two input XOR gate



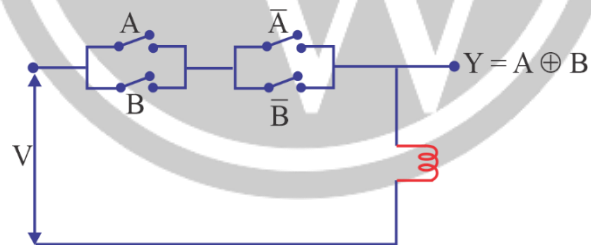
Truth table for 2-input XOR gate

Input		Output
A	B	$Y = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

XOR gate using AND, OR and NOT gate



Switching diagram of XOR gate



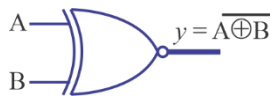
Truth Table:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

$$\begin{aligned}
 Y &= (A + B)(\bar{A} + \bar{B}) \\
 &= \bar{A}B + A\bar{B} \\
 &= A \oplus B
 \end{aligned}$$

1.2.4. X-NOR gate:

Symbol for two input X-NOR gate



Truth table for 2-input X-NOR gate

Input		Output
A	B	$Y = A \odot B$
0	0	1
0	1	0
1	0	0
1	1	1

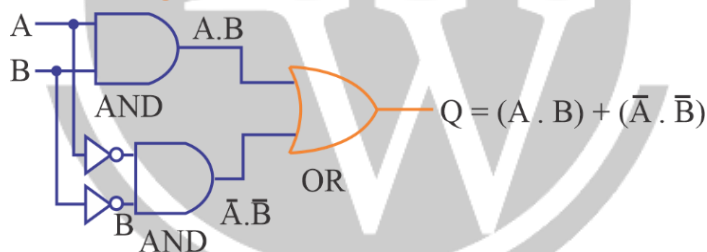
Boolean expression for EX-NOR gate is $Y = \overline{A \oplus B}$

Apply De-Morgan's theorem:

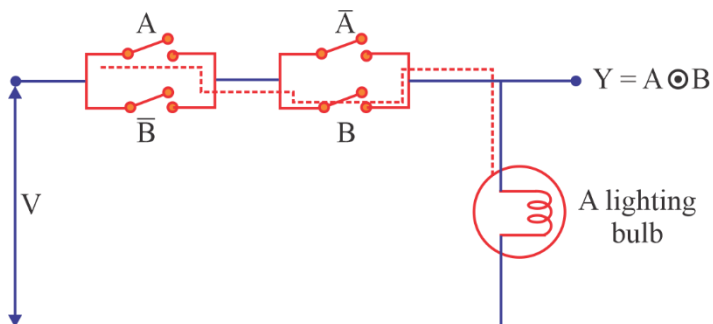
$$\overline{A \oplus B} = \overline{AB + \bar{A}\bar{B}} = \overline{AB} \cdot \overline{\bar{A}\bar{B}} = (A + \bar{B})(\bar{A} + B) = AB + \bar{A}\bar{B}$$

The output of a two input EX-NOR gate is logic '1' when the inputs are same and a logic '0' when they are different.

X-NOR gate using AND OR and NOT gate



Switching Diagram of X-NOR gate



$$\begin{aligned} Y &= (A + \bar{B})(\bar{A} + B) \\ &= AB + \bar{A}\bar{B} \\ &= A \oplus B \end{aligned}$$

Truth Table:

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

1.3. Alternate Logic Gate Representation

Logic	Normal symbol	Alternate symbol
NOT	$A \rightarrow \neg A$	$A \rightarrow \neg A$
AND	$A, B \rightarrow AB$	$A, B \rightarrow \overline{\overline{A} + \overline{B}} = AB$
OR	$A, B \rightarrow A + B$	$A, B \rightarrow \overline{\overline{A} \cdot \overline{B}} = A + B$
NAND	$A, B \rightarrow \overline{A \cdot B}$	$A, B \rightarrow \overline{\overline{A} + \overline{B}} = \overline{A \cdot B}$
NOR	$A, B \rightarrow \overline{A + B}$	$A, B \rightarrow \overline{\overline{A} \cdot \overline{B}} = \overline{A + B}$

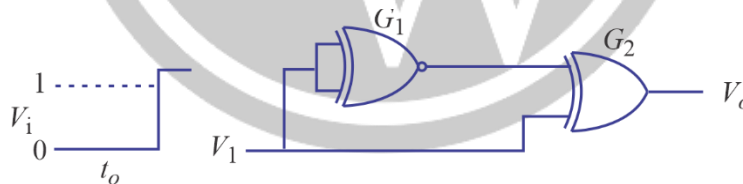
Example: In the following circuit, find the output Z?

Solution: From the given circuit, we can observe that input to last XNOR is same, so, the XNOR output is given by (let input is X)

$$Z = X \cdot X + \overline{X} \cdot \overline{X} = X + \overline{X} = 1$$

i.e. the output will be high [logic 1] irrespective of the inputs A and B.

Example: The gate G_1 and G_2 in figure shown below have propagation delays of 10ns and 20ns respectively.



If input V_i makes an abrupt change from logic 0 to 1 at $t = t_0$, then find the output waveform V_o ?

Here, $t_1 = t_0 + 10 \text{ ns}$, $t_2 = t_1 + 10 \text{ ns}$, $t_3 = t_2 + 10 \text{ ns}$.

Solution: Let the output of $G_1 = X$

The output waveform will be as shown in figure below.

